

THAPELO WANA

Software Engineer · Computer Science & Electronics
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PROFESSIONAL SUMMARY

Final-year BSc Computer Science and Electronics student at North-West University, maintaining a 73%+ academic average across software, electronics, and AI modules. Skilled in building full-stack web applications end-to-end -from relational database design and REST API development to React front-ends with authentication and role-based routing. Experienced in leading teams through complete software development lifecycles, with additional hands-on exposure to machine learning (YOLOv8, K-Means, NLP), analogue electronics, and signals processing. Seeking a graduate or intern software engineering role to grow within a high-performance engineering team.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES2022+), Python, Java, C++, SQL
Frontend: React 18, React Native, Vite, TanStack Query, Tailwind CSS, React Router, zod, react-hook-form
Backend & Databases: Node.js, Express, Supabase (Auth, PostgreSQL, Edge Functions), Firebase, MySQL
AI & ML: YOLOv8 object detection, K-Means clustering, NLP experimentation (Qwen, DistilGPT2), MATLAB
Engineering: REST API design, OOP, UML, Git/GitHub (branching, PR workflows), debugging & testing
Electronics: Analogue circuit design, op-amp configurations, signal processing, MATLAB simulation

PROJECTS

Clinic Appointment Scheduling System (CASS) – NWU CMPG224 | Lead Developer, 3-person development team | 2025

- Designed and built a full-stack appointment platform using React 18, TypeScript, and Supabase, supporting three distinct user roles (patient, doctor, admin) with role-based UI routing and access control implemented at both client and database level.
- Implemented the complete appointment booking workflow: doctor discovery with specialisation and fee filters, date-constraint enforcement, time-slot selection, and a server-side conflict validator that prevents double-bookings across all available slots.
- Built end-to-end authentication including session lifecycle management, profile hydration on login, and protected routes, strengthening role-based access across patient, doctor, and admin views.
- Developed a notification pipeline using Supabase Edge Functions and the Resend API, with per-notification status tracking (pending / sent / failed / skipped) and error recovery logic.

Stack: TypeScript, React 18, Vite, React Router, TanStack Query, Tailwind CSS, Supabase, Resend API, zod, react-hook-form

CineBook – Online Movie Ticket Booking System | NWU CMPG311 – Project Manager, 8-person team | 2025

- Led an 8-person team through the full SDLC -covering requirements analysis, sprint planning, task allocation, and delivery milestones from kick-off through final demo and project report.
- Implemented core React components (movie browsing, seat selection, booking confirmation) and 6+ RESTful API routes in Node.js/Express covering listings, showtimes, seat availability, booking creation, and JWT-based authentication.
- Designed the relational database schema in MySQL and contributed Oracle SQL artefacts including views, triggers, stored procedures, and 10 analytical queries for the academic submission.
- Established the GitHub repository structure, branching strategy, and pull request workflow, enabling parallel development across 8 contributors throughout the project lifecycle.
- Produced Phase 1 system design documentation including ER diagrams and data-flow models that served as the shared technical blueprint for the entire team.

Stack: React.js, Node.js, Express, MySQL, Oracle SQL, JWT, Git/GitHub

GitHub Portfolio – AI, Algorithms & Software Projects | github.com/Naxisbeast | 2024–2025

- Object detection pipeline using YOLOv8 and Byte Track on custom footage; experimented with confidence thresholds and evaluated tracking accuracy across frames.
- Network topology clustering using K-Means in Python; analysed cluster stability across different k values and visualised decision boundaries.
- NLP comparison study evaluating ELIZA (rule-based) against transformer-based models (Qwen, DistilGPT2) on conversational response quality and task completion.
- Algorithmic problem-solving repository covering BFS, DFS, A* search, and First-Order Logic resolution with annotated implementations and complexity notes.

EXPERIENCE

Student Assistant – Physics Department | North-West University, Mahikeng | 2025 – Present

- Support electronics and signals lab sessions for approximately 30 students per session, covering analogue circuit setups, op-amp configurations, signal processing practicals, and oscilloscope measurements.
- Diagnose and resolve student circuit faults in real time, explaining root causes clearly to build understanding rather than just correcting errors.
- Collaborate with academic staff to keep sessions on schedule and adapt explanations to students at different skill levels.
- Reinforce theory-to-practice connections across topics including Fourier analysis, RC circuits, and feedback amplifier behaviour.

VIRTUAL EXPERIENCE & CERTIFICATIONS

Software Engineering Virtual Experience | Electronic Arts (EA) via Forage | Nov 2025

- Authored a formal feature proposal covering technical scope and cross-functional stakeholder value across four functional areas.
- Designed a UML class hierarchy and implemented corresponding C++ header files for a core game module.
- Identified a performance bottleneck and resolved it by substituting an $O(\log n)$ data structure, reducing computational overhead on the critical path.

ICG Technology Software Development Job Simulation | Citi via Forage | Nov 2025

- Built a Java tool to visualise live stock market risk data, enabling sub-second rendering of metrics previously handled via static reports.
- Modelled a loan management workflow as a UML state diagram, documenting 12+ transition paths and surfacing two process inefficiencies for developer handoff.
- Delivered a structured recommendation report evaluating logistic regression, gradient boosting, and neural networks for credit risk assessment, with reasoned model selection.

EDUCATION

BSc Computer Science and Electronics | North-West University, Mahikeng Campus | Feb 2024 – Nov 2026 (Expected)

- Academic average: 73%+ maintained across technical and engineering modules throughout the programme.
- Relevant coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Software Engineering, Artificial Intelligence, Advanced Analogue Electronics, Signals & Systems, Circuit Design.

CORE COMPETENCIES

Full-stack system design · Technical team leadership · Analytical problem-solving · Clear technical communication · Debugging under pressure · Rapid skill acquisition · Structured documentation

AVAILABILITY & WORK AUTHORISATION

South African citizen. Academic average 73%+. Available full-time from graduation (November 2026); available during university vacation periods (June/July 2026). Open to in-person or hybrid arrangements in Johannesburg, Pretoria, or remotely.